

Basics of FE-Analysis

BK10A6400

Lecture 6: Numerical integration, simple examples

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October 5, 2026

Numerical integration

- Basis of numerical integration (MathWorld), for example.
- To find an approximate solution to a definite integral
$$\int_{-1}^1 f(x) dx \approx \sum_{i=1}^{nip} f(x_i) w_i.$$
- Needed to evaluate area or volume integral to get $\overline{K}$
- Analytically before - Why make it numerically?
 - Faster (for complicated element)
 - Analytical solution is not always possible
 - Specific numerical integration schemes are needed to get rid of locking phenomena
- Gaussian quadratures are used because a polynomial of degree $p = 2nip - 1$ can be integrated exactly with nip integration points

$$W_{int} = \frac{1}{2} \int_V \boldsymbol{\sigma}^T \boldsymbol{\varepsilon} dV = \frac{1}{2} \mathbf{u}^T \underbrace{\int_V \mathbf{B}^T \mathbf{D} \mathbf{B} dV}_{\overline{K}} \mathbf{u} \quad (1)$$

$$W_{int} = \frac{1}{2} l_z \int_A \boldsymbol{\sigma}^T \boldsymbol{\varepsilon} dA = \frac{1}{2} \mathbf{u}^T l_z \underbrace{\int_A \mathbf{B}^T \mathbf{D} \mathbf{B} dA}_{\overline{K}} \mathbf{u} \quad (2)$$

Matlab class assignment: Numerical integration

Use Gauss quadrature integration scheme to estimate function $A(x) = \int_a^b f(x)dx$ where $f(x) = x^2$ and $a = 0, b = 3$.

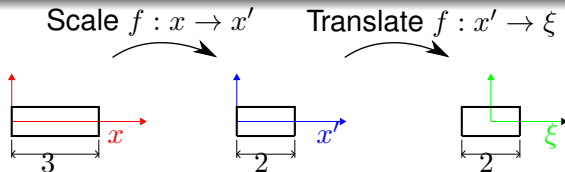
$$A = \int_a^b f(x)dx = \int_0^3 x^2 dx \quad (3)$$

Transformation: $x \rightarrow \xi$

$$x = \frac{b}{2}(\xi + 1) \quad ; dx = \frac{b}{2} d\xi \quad (4)$$

$$A = \int_{-1}^1 f(\xi) \frac{b}{2} d\xi \approx \sum_{i=1}^{nip} f(\xi_i) w_i \frac{b}{2} \quad (5)$$

Matlab class assignment: transformation $x \rightarrow \xi$ step by step



Scaling $x \rightarrow x'$ (from coordinates $0 \dots 3 \rightarrow 0 \dots 2$) :

$$x' = \frac{2}{3}x \quad ; \quad x'(x=0) = 0; x'(x=3) = 2 \quad (6)$$

Translating $x' \rightarrow \xi$ (from coordinates $0 \dots 2 \rightarrow -1 \dots 1$) :

$$\xi = x' - 1 \quad ; \quad \xi(x'=0) = -1; \xi(x'=2) = 1 \quad (7)$$

Substituting x' , final eq:

$$\xi = \frac{2}{3}x - 1 \quad ; \quad \xi(x=0) = -1; \xi(x=3) = 1 \quad (8)$$

Matlab class assignment: transformation $x \rightarrow \xi$ step by step

Now x can be solved from Eq. (8):

$$x = \frac{3}{2}(\xi + 1) \quad (9)$$

and differential dx :

$$\frac{dx}{d\xi} = \frac{3}{2} \rightarrow dx = \frac{3}{2}d\xi \quad (10)$$

Now the area $A(x)$ can be written

$$A = \int_0^3 x^2 dx = \int_{-1}^1 \underbrace{\left(\frac{3}{2}(\xi + 1)\right)^2}_{f(\xi)} \underbrace{\frac{3}{2} d\xi}_{dx} \quad (11)$$

A number of integration points for an accurate solution. The degree of a polynomial of the integrand $p = 2$.

$$nip = \frac{(p + 1)}{2} = 1.5 \quad (12)$$

so two integration points are needed. For Gaussian quadrature integration, integration points and weights can be found in mathematics handbooks.

Matlab class assignment: Numerical integration in Matlab

Two point Gauss quadrature formula is exact for polynomials up to degree 2:

```
xiv=[-0.57735026918963,0.57735026918963];  
wxiv=[1,1];
```

You need to write function where is your integrand $f(x) = x^2$ and then main code where integration is done:

```
f=0;  
for i=1:nip,  
    f=f+wxiv(i)*fdxi(xiv(i));  
end
```

Numerical integration in 2D and 3D

2D:

$$\int_{-1}^1 \int_{-1}^1 f(\xi, \eta) d\xi d\eta \approx \sum_{j=1}^{nip_{\eta}} \sum_{i=1}^{nip_{\xi}} f(\xi_i, \eta_j) w_{\xi_i} w_{\eta_j} \quad (13)$$

3D:

$$\int_{-1}^1 \int_{-1}^1 \int_{-1}^1 f(\xi, \eta, \zeta) d\xi d\eta d\zeta \approx \sum_{k=1}^{nip_{\zeta}} \sum_{j=1}^{nip_{\eta}} \sum_{i=1}^{nip_{\xi}} f(\xi_i, \eta_j, \zeta_k) w_{\xi_i} w_{\eta_j} w_{\zeta_k} \quad (14)$$

Shear locking and selective reduced integration

To get rid of shear locking (of bilinear plate element), you should make separated integration for energies related to shear and normal deformations.

$$U = U_n + U_s \quad (15)$$

$$U_n = \frac{1}{2} l_z \int_V \boldsymbol{\sigma}_n^T \boldsymbol{\varepsilon}_n \det(\mathbf{J}_e) d\xi = \frac{1}{2} l_z \int_V \boldsymbol{\varepsilon}_n^T \mathbf{D}_n \boldsymbol{\varepsilon}_n \det(\mathbf{J}_e) d\xi \quad (16)$$

$$U_s = \frac{1}{2} l_z \int_V \sigma_{xy} \gamma_{xy} \det(\mathbf{J}_e) d\xi = \frac{1}{2} l_z \int_V D_s \gamma_{xy}^2 \det(\mathbf{J}_e) d\xi \quad (17)$$